

ERAFIT Platform Architecture & Features

Comprehensive technical reference detailing the frontend ecosystem, backend services, relational data models, clinical telemetry, multilingual voice AI engine, and full feature capabilities of the ERAFIT Platform.

APPLICATION TYPE	PRIMARY STACK	DATABASE & ORM	DOCUMENT VERSION
Next.js 16 Turbopack Web App	React 19 + TypeScript + Prisma	SQLite + Prisma ORM 6.19	v1.0 (Production Build)

1 Complete Technology Stack (Frontend & Backend)

⚡ Frontend Ecosystem

Client-Side

- **Next.js 16.3.1 (App Router + Turbopack)**
Server & Client Components, file-based routing, nested layouts, optimized font & image delivery.
- **React 19.2.8 & React DOM 19**
Concurrent rendering, modern action hooks, Context API state orchestration, resilient rendering.
- **TypeScript 5.x**
Strict end-to-end type safety, unified data contracts between UI views and backend Route Handlers.
- **Tailwind CSS v4 & PostCSS**
Utility-first dark/neon glassmorphism UI, custom micro-interactions, responsive mobile-first layouts.
- **Lucide React Icons (v1.31.0)**
Crisp vector iconography across athletic dashboards, telemetry meters, and navigation bars.
- **Web Speech API (Recognition + Synthesis)**
Zero-dependency browser native voice AI with multi-language voice dictation and audio feedback.
- **HTML5 MediaDevices & Canvas API**
Live camera frame capture for real-time AI food scanning and visual barcode/dish estimation.

🛡 Backend & Infrastructure

Server-Side

- **Node.js LTS (v24.x) Runtime**
High-performance async server runtime executing Next.js Route Handlers and telemetry tasks.
- **Prisma ORM 6.19.3**
Type-safe schema modeling, relational querying, automated migrations, and connection pooling.
- **SQLite Database Engine (dev.db)**
Embedded ACID-compliant relational storage for users, clinical biomarkers, food logs, and workouts.
- **Cookie Session Auth & 3-Tier RBAC**
Role-Based Access Control enforcing strict separation: Athlete (USER), Coach (MONITOR), Overseer (ADMIN).
- **Sharp & PNGJS Image Processing**
Server-side high-throughput image manipulation, compression, and visual telemetry data parsing.
- **XLSX & CSV Data Pipelines**
High-speed dataset stream loaders powering the 5,000+ Indian & USDA clinical food database.
- **NLP Dialect & Phonetic Parser**
Custom regional language parser for English, Tamil (தமிழ்), Hindi (हिंदी), and Malayalam (മലയാളം).

💡 **Architecture Highlights:** ERAFIT utilizes a hybrid Server/Client component paradigm with Next.js Turbopack, maintaining zero external dependencies for voice AI via the Web Speech API and instant sub-millisecond local SQLite database transactions via Prisma ORM.

2 Relational Database Models (Prisma Schema)

ERAFIT maintains a comprehensive relational schema in Prisma ORM mapped to an SQLite database, supporting clinical health markers, multi-role RBAC, 5,000+ food items, and live exercise telemetry.

MODEL NAME	KEY ATTRIBUTES & FIELDS	DOMAIN PURPOSE & RELATIONSHIPS
User	id, name, email, role (USER/MONITOR/ADMIN), gender, age, heightCm, weightKg, goal, bloodType, trainerSync	Central identity & athlete profile linked to all telemetry, medical reports, food logs, and coach assignments.
MonitorProfile	userId, monitorId (e.g. ERA-MON-8942), specialization, experienceYears, status	Profiles for certified sports medicine trainers, clinical nutritionists, and workout monitors.
UserMonitorAssignment	userId, monitorId, assignedAt, status (ACTIVE/PAUSED), assignedBy	M:N relational mapping connecting athletes with their dedicated clinical/fitness coach.
AssignmentHistory	userId, userName, monitorId, monitorName, action, performedBy, timestamp	Complete chronological audit trail of all athlete-to-monitor assignments and status updates.
MonitorNote	userId, monitorId, title, content, category (WORKOUT/DIET/MEDICAL), isTask, completed	Clinical observations, coach feedback notes, and assigned trainee action items.
AssignedWorkoutPlan	userId, title, notes, exercises (JSON payload: sets/ reps/weight), assignedByName	Prescription-based customized workout regimens dispatched directly by monitors to trainees.
AssignedDietPlan	userId, title, targetCalories, targetProtein, targetCarbs, targetFat, meals (JSON)	Clinical nutrition regimens customized per user based on health parameters and dietary goals.
HealthReport & BloodParameter	reportDate, packageld, testName, value, unit, referenceLow, referenceHigh, status (NORMAL/LOW/HIGH/ATTENTION)	Thyrocare / Aarogyam laboratory blood tests (Lipid, Liver, Thyroid, Vitamin, CBC) with normal/high alerts.
DXAReport & DXAParameter	reportDate, scanTypeId, metricName (Body Fat, Lean Mass, BMD, T-Score), region (Arms, Legs, Trunk)	Clinical Dual-energy X-ray Absorptiometry bone density & segmental body composition tracking.
Food & FoodLog	name, calories, protein, carbs, fat, fiber, sugar, sodium, vegetarian, vegan, mealType (Breakfast/Lunch/Dinner/Snack)	5,000+ item nutrition dictionary + daily timestamped breakfast, lunch, dinner, snack user logs.
Workout, WorkoutSet & Exercise	name, startTime, durationMin, setNumber, reps, weight, completed	Live workout tracking engine storing barbell telemetry, weight progression, and volume metrics.
WaterLog & HealthMetric	amountMl, metric (Weight/BodyFat), value, logDate	Hydration logging and longitudinal biometric evolution metrics.
AIConversation & AuditLog	role, content, timestamp, actorId, actorRole, action, previousValue, newValue	Persistent AI Coach dialog memory and enterprise audit logging for security compliance.

3 Core Modules & Comprehensive Feature Breakdown

14 complete production modules implemented across the ERAFIT application spanning clinical diagnostics, multilingual voice AI, live telemetry, and multi-tier coaching.

1. Auth & Multi-Tier RBAC

/auth/*

Full auth workflow with separate registration portals for Athletes, Certified Monitors, and Platform Admins. Includes role verification, cookie session persistence, and protected route middleware.

2. Main Fitness & Telemetry Dashboard

/

Unified overview displaying real-time daily calorie targets, macronutrient split (Protein, Carbs, Fats), circular progress gauges, water intake tracker, quick workout launcher, and trainer sync status.

3. Multilingual AI Nutrition Voice Assistant

/nutrition

AI voice agent with natural speech input. Automatically detects English, Tamil (தமிழ்), Hindi (हिंदी), and Malayalam (മലയാളം), maps colloquial queries to clinical items, and speaks spoken answers aloud.

4. Live Camera AI Food Scanner

/scanner

Interactive camera scanner using WebRTC video feed with a targeting reticle. Captures live meal snapshots, performs visual dish identification, and provides instant macro estimates with one-tap logging.

5. 5,000+ Food Clinical Nutrition Engine

FIT_ERA_5000

Comprehensive database combining Indian Food Composition Tables (IFCT) and USDA databases. Searchable by meal category, vegetarian/vegan filters, with micronutrients (Iron, Calcium, Vitamin C).

6. Clinical Blood Biomarker Panel

/health/blood-panel

Integrated with Thyrocare Aarogyam clinical packages (60+ biomarkers). Automatically flags parameters as Normal, Low, High, or Attention for Lipid, Liver, Kidney, Thyroid, and Iron panels.

7. DXA Body Composition & Bone Density

/health/dxa-results

Dual-energy X-ray Absorptiometry report parser tracking regional body fat %, lean mass distribution (arms, legs, trunk), visceral adipose tissue, Bone Mineral Density (BMD), and clinical T-Scores.

8. Workout Routines & Active Drawer

/workouts

Extensive exercise library categorized by muscle groups (Chest, Back, Legs, Arms). Features a global persistent floating drawer for active workout sessions with rest timers, reps, and weight logging.

9. Live Barbell & Bench Telemetry

/workout-tracker/*

Real-time telemetry tracker for barbell movements including bench press, squat, and deadlift. Records per-set tonnage, RPE (Rate of Perceived Exertion), set completion checkmarks, and progression curves.

10. Body Composition & TDEE Planner

/body-analysis

Scientific calculator estimating Basal Metabolic Rate (BMR), Total Daily Energy Expenditure (TDEE), target caloric surplus/deficit, and custom macro ratios tailored to muscle gain, fat loss, or maintenance.

11. Monitor & Trainer Clinical Portal

/monitor/*

Dedicated coach workspace: inspect assigned trainees, review daily telemetry logs, view blood test alerts, write clinical notes/tasks, and dispatch custom workout and meal prescriptions.

12. Admin Control & Assignment Hub

/admin/*

Platform governance suite: oversee all system users, verify monitor credentials, dynamically assign/reassign athletes to coaches, and review comprehensive audit history logs.

13. Conversational AI Fitness Coach

/ai-coach

AI chatbot trained on sports science and clinical nutrition principles. Maintains conversation history, provides form corrections, dietary tips, and recovery protocols.

14. Wearable Devices & Hardware Hub

/devices

Telemetry interface for smartwatches, continuous glucose monitors (CGM), heart rate monitors, and fitness trackers with real-time biometric synchronization.

4 RESTful API Endpoint Matrix

Core API route handlers powering user telemetry, clinical health data ingestion, diet prescriptions, and multi-tier RBAC.

METHOD	ENDPOINT ROUTE	REQUEST / QUERY PARAMETERS	FUNCTIONAL DESCRIPTION
POST	/api/auth/login	email, password, expectedRole	Authenticates user credentials and sets role session cookie.
POST	/api/auth/register	name, email, password, biometrics	Registers new athlete profile with baseline metrics.
GET	/api/auth/me	- (Session Cookie)	Fetches active authenticated user and role context.
GET	/api/foods	search, category, veg, vegan	Queries 5,000+ nutrition items with filters and pagination.
POST	/api/nutrition/voice-agent	query, lang (en/ta/hi/ml)	NLP query parsing, multi-lingual translation & macro calculation.
POST	/api/nutrition	foodId, mealType, quantity	Logs a consumed food item to user's daily diary.
GET	/api/health/blood-panel	userId, packageId	Returns clinical biomarker analysis with normal/attention ranges.
POST	/api/health/upload-blood	parameters[], reportDate, lab	Stores blood report data with OCR or verified manual values.
GET	/api/health/dxa	userId	Retrieves latest DXA body composition & bone scan telemetry.
POST	/api/workouts	name, duration, sets[]	Saves completed workout session with exercise set weights & reps.
POST	/api/water	amountMl	Logs incremental water consumption in milliliters.
GET	/api/monitor/users	monitorId	Fetches all athletes assigned to the authenticated monitor.
POST	/api/monitor/users/[id]/edit	workoutPlan, dietPlan, notes	Dispatches custom workout and diet plans to assigned athlete.
POST	/api/admin/assignments	userId, monitorId, action	Admin endpoint to assign, reassign, or revoke trainer links.
POST	/api/ai-coach	message, history[]	Dispatches conversational query to AI Fitness Coach.



Deployment Ready: ERAFIT is engineered for scalable deployment on Next.js edge runtimes or Node.js server environments with seamless migration paths from SQLite to PostgreSQL / MySQL via Prisma schema configuration.